3.3 Halogen Compounds

Question Paper

Course	CIE A Level Chemistry
Section	3. Organic Chemistry
Topic	3.3 Halogen Compounds
Difficulty	Hard

Time allowed: 60

Score: /47

Percentage: /100

Question 1a

This question is about halogenoalkanes.

A student investigates the rate of reaction of six halogenoalkanes using the following method.

1. Mix ethanol with six drops of halogenoalkane.
2. Warm the mixture in a water bath at 50 °C.
3. Add silver nitrate solution.
4. Record the time taken for the precipitate to form.

Table 1.1 shows the student's results.

Table 1.1

Halogenoalkane	**Time taken for the precipitate to form (s) at 50 °C**
$CH_3CH_2CH_2CH_2CH_2Cl$	265
$CH_3CH_2CH_2CH_2CH_2Br$	152
$CH_3CH_2CH_2CH_2CH_2I$	65
$CH_3CH_2CH_2CHBrCH_3$	88
$(CH_3)_2CHCHClCH_3$	190
$CH_3CH_2C(CH_3)_2Cl$	88

i)
Suggest how the student could improve the method.

[3]

ii)
Other than precipitation, state what type of reaction is occurring in this method.

[1]

[4 marks]

Question 1b

Using Table 1.1, describe and explain **two** factors that influence the rate of this type of reaction in halogenoalkanes.

[4 marks]

Question 1c

A fresh sample of aqueous 1-bromopentane reacts with an aqueous solution of sodium hydroxide.

i)

Fully identify the mechanism for this reaction and state the number of steps in the mechanism.

[1]

ii)

One of the products contains bromine. Name this product.

[1]

iii)

The reaction conditions are changed resulting in the formation of three products, including the product identified in part **(c)(ii)**.

Suggest the change that is made to the conditions.

[1]

[3 marks]

Question 1d

1-chloropentane was prepared in the laboratory by the reaction of an alcohol and thionyl chloride, $SOCl_2$.

Two additional products are also produced in this reaction.

i)

Give the equation, including state symbols, for the preparation of 1-chloropentane.

[2]

ii)

Explain **one** safety precaution necessary for carrying out this reaction. Justify your answer.

[2]

[4 marks]

Question 2a

This question is about the reactions of halogenoalkanes.

2-bromo-2-methylpropane is refluxed with ethanolic potassium hydroxide, KOH. The result is a mixture of products containing roughly a 4 : 1 ratio of alkene to alcohol.

Complete Table 2.1 to name the organic products and identify the mechanisms involved.

Table 2.1

Organic product	Name	Mechanism involved
Alkene		
Alcohol		

[2 marks]

Question 2b

Suggest why there is a mixture of an alkene and an alcohol produced in the reaction outlined in part **(a)**.

[2 marks]

Question 2c

An iodoalkane is prepared by the reaction of red phosphorus, P, with iodine, I_2, to produce phosphorus(III) iodide, PI_3. This then is reacted with an alcohol to produce the iodoalkane and phosphorous acid, H_3PO_3.

Give the equation for the reaction between butan-2-ol and phosphorus(III) iodide.

[2 marks]

Question 2d

During a series of reactions to produce a bromoalkane, a 50 : 50 mix of sulfuric acid and water is reacted with potassium bromide to produce hydrogen bromide. The mixture is kept very cool.

Explain why sulfuric acid can not be used in the preparation of an iodoalkane

[1 mark]

Question 3a

Compound **G** is a chloroalkane that can undergo two different reactions as outlined in Fig. 3.1.

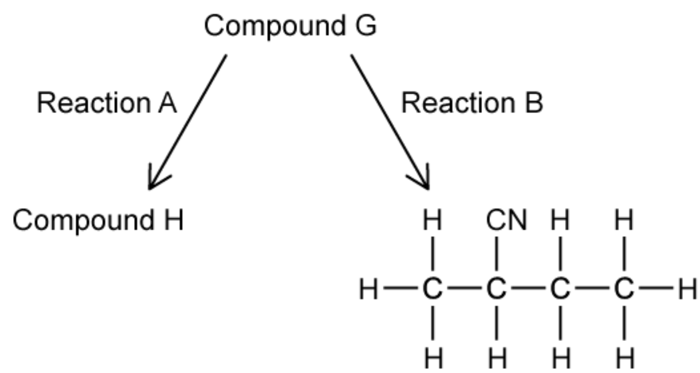


Fig. 3.1

Compound **H** is an alkene which does not exhibit geometric isomerism.

Draw the skeletal structures of compounds **G** and **H**.

[2 marks]

Question 3b

State the reagents and conditions that will be required for reaction **A** in Fig. 3.1.

[2 marks]

Question 3c

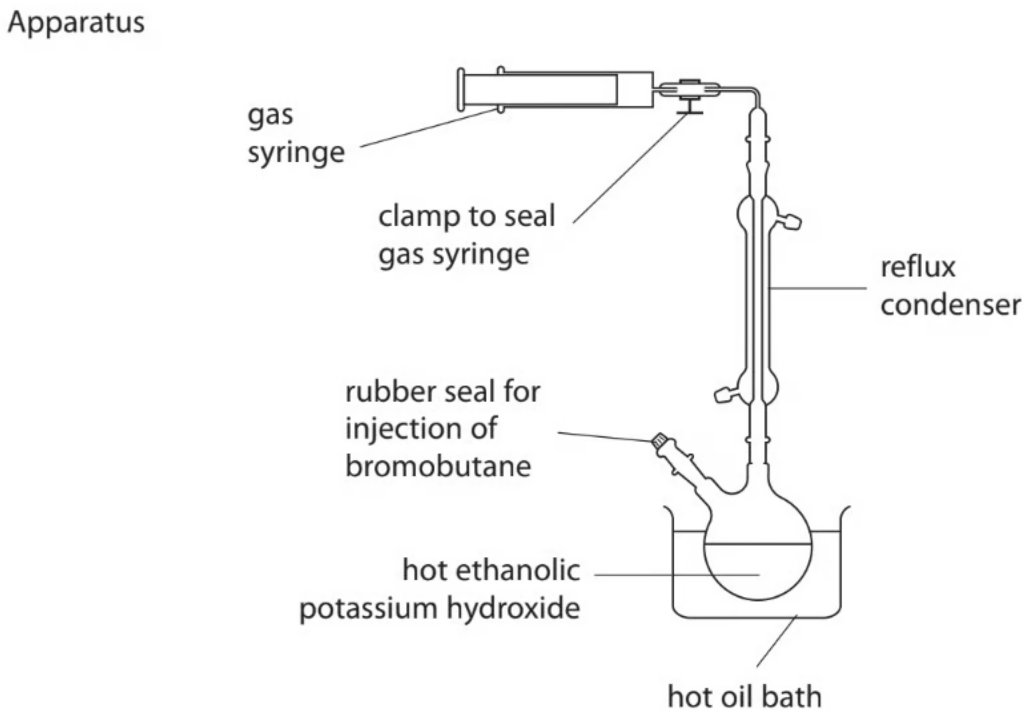
Name and draw the mechanism for reaction **B** in Fig 3.1. Include all charges, partial charges, lone pairs and curly arrows.

[4 marks]

Question 4a

Bromobutanes react with hot ethanolic potassium hydroxide solution to produce gaseous butenes.

$$C_4H_9Br + OH^- \rightarrow C_4H_8 + Br^- + H_2O$$



Procedure

- 0.0080 mol of liquid 1-bromobutane was injected into a round bottom flask containing hot ethanolic potassium hydroxide.
- After the reaction, the syringe was sealed using a clamp.
- The syringe was then removed from the apparatus and allowed to cool to 25°C.

Result

- The final volume of but-1-ene collected was 22.0 cm^3.

State the purpose of the condenser.

[1 mark]

Question 4b

Describe a chemical test and the expected observation to identify the functional group of the gas in the syringe.

[3 marks]

Question 4c

Calculate the percentage of 1-bromobutane which was converted to but-1-ene.

[2 marks]

Question 4d

Before cooling, the volume of but-1-ene in the gas syringe was 24.0 cm^3.

Assuming a constant pressure and that no but-1-ene is lost from the gas syringe during cooling, calculate the temperature of the gas in the syringe.

Temperature = ...

[2 marks]

Question 4e

i)

Another compound formed from 1-bromobutane under these conditions is butan-1-ol.

Fully identify the type of reaction taking place to form butan-1-ol.

[1]

ii)

Draw the mechanism for the reaction of 1-bromobutane with hydroxide ions to form butan-1-ol. Include all charges, partial charges, lone pairs and curly arrows.

[3]

iii)

Describe a chemical test and the expected observation(s) to confirm the presence of the functional group in butan-1-ol.

[2]

[6 marks]

Question 4f

Isomeric alkene molecules are formed in the elimination reaction of 2-bromobutane.

Draw the displayed formulae of the isomers formed during this reaction.

[3 marks]

Model Answers: Hard

1a

a)

i) Alterations the student could make to the method are:

- Add the same volume / a specified volume of ethanol to each test tube / reaction
  AND
  Add the same volume / a specified volume of silver nitrate to each test tube / reaction; [1 mark]
- Use measured volumes of the halogenoalkanes (rather than 6 drops); [1 mark]
- Ensure the silver nitrate is warmed to 50 °C before adding
  OR
  Ensure that all the reactants are at a constant temperature before mixing them; [1 mark]

ii) The reaction occurring in this method is:

- Hydrolysis (in aqueous conditions)
  OR
  Nucleophilic substitution ; [1 mark]

[Total: 4 marks]

- In order to score all the marks for part (i), you have to give all the alterations required to make this method far more accurate
- The addition of the same volume of ethanol and silver nitrate is the same idea
  - This is why both comments are needed to score the mark
- Ensuring all reactants are at the same temperature before mixing is a crucial step and will ensure the temperature will remain as close to 50 °C during the reaction

1b

b) Description and explanation of the factors that influence the rate of this type of reaction in halogenoalkanes:

Factor 1 - halogen

- Chloroalkanes react the slowest
  OR
  Iodoalkanes react the fastest
  OR
  Chloroalkanes react the fastest, then bromoalkanes with iodoalkanes reacting the slowest; [1 mark]
- (Because) the bond strength of C-Cl > C-Br > C-I bond
  OR
  (Because) the C-I bond is hydrolysed faster than C-Br or C-Cl bonds / the C-I bond is hydrolysed faster than C-Br which is hydrolysed faster than the C-Cl bond
  OR
  (Because) chlorine has a higher electronegativity than bromine or iodine / chlorine has a higher electronegativity than bromine which has a higher electronegativity than iodine
  OR
  (Because) the C-Cl bond has a greater dipole than C-Br or C-Cl bonds / the C-Cl bond has a greater dipole than the C-Br bond which has a great dipole than the C-Cl bond;
  [1 mark]

Factor 2: - structure

- Branched-chain halogenoalkanes react faster than straight-chain halogenoalkanes
  OR
  Tertiary halogenoalkanes react faster than secondary **AND** secondary halogenoalkanes react faster than primary; [1 mark]
- (Because) branched-chain halogenoalkanes form more stable carbocations
  OR
  (Because) tertiary carbocations are more stable than secondary carbocations **AND** secondary carbocations are more stable than primary carbocations;
  [1 mark]

[Total: 4 marks]

- You are asked to describe two factors that influence the rate of reaction
  - The results table shows that a variety of halogenoalkanes are being used:

    - Primary, secondary and tertiary
    - Chloroalkanes, bromoalkanes and iodoalkanes
    - Straight-chain and branched-chain
- You can use the results table to identify the relevant trends:
  - Primary halogenoalkanes react slower than secondary, which react slower than tertiary halogenoalkanes
  - Chloroalkanes react slower than bromoalkanes, which react slower than iodoalkanes
  - Straight-chain halogenoalkanes react slower than branched-chain halogenoalkanes
- Once you have used the results to describe two factors, you then have to explain them
  - **Careful:** The primary / secondary / tertiary halogenoalkanes and straight-chain / branched-chain halogenoalkanes are essentially the same points with the same explanation
  - This means that they are covered by the same marking points in the mark scheme
- The explanation for the chloroalkanes / bromoalkanes / iodoalkanes can talk about:
  - The strength of the carbon-halogen / C-X bond
  - The hydrolysis of the carbon-halogen / C-X bond
  - The electronegativity of the halogen atom involved in the carbon-halogen / C-X bond
  - The strength of the dipole formed in the carbon-halogen / C-X bond
- The explanation for the straight-chain / branched-chain halogenoalkanes must talk about the stability of the intermediate carbocations that form

1c

c)

i) The mechanism for the reaction of aqueous 1-bromopentane reacts with an aqueous solution of sodium hydroxide and number of mechanism steps are:

- Mechanism = S_N2
  AND
  Number of steps = 1; [1 mark]

ii) The product that contains bromine is:

- Sodium bromide; [1 mark]

ii) The change to the conditions is:

- Change the sodium hydroxide from aqueous to ethanolic / alcoholic; [1 mark]

[Total: 2 marks]

- The following reaction occurs:
  - $CH_3CH_2CH_2CH_2CH_2Br + NaOH \rightarrow CH_3CH_2CH_2CH_2CH_2OH + NaBr$
- The hydroxide ion / OH^- acts as the nucleophile attacking the δ+ carbon of the C-Br bond and forming a new C-OH bond
  - At the same time, the C-Br bond breaks heterolytically to form the bromide ion
  - The bromide ion is attracted to the sodium ion and forms sodium bromide
- This is a one-step reaction **AND** the 1-bromopentane is a primary halogenoalkane
  - Therefore, this is an S_N2 mechanism
  - You will not gain the mark for writing "nucleophilic substitution" as the question asks you to fully identify the mechanism, which means the examiner wants to know if it is S_N1 or S_N2
  - **Careful:** S_N1 mechanisms have two steps and S_N2 mechanisms have one step
- Nucleophilic substitution generally occurs when the sodium hydroxide is aqueous and elimination occurs when the sodium hydroxide is alcoholic / ethanolic
  - **Careful:** You are asked for the change to the conditions so you need to make sure that you say what you are changing, i.e. the original conditions **AND** the new conditions

1d

d)

i) The equation for the reaction of 1-chloropentane and thionyl chloride, $SOCl_2$ is:

$CH_3CH_2CH_2CH_2CH_2OH$ (l) + $SOCl_2$ (l)	→	$CH_3CH_2CH_2CH_2CH_2Cl$ (l) + HCl (g) + SO_2 (g)

- Correct chemical formulae; [1 mark]
- Correct state symbols; [1 mark]

ii) A safety precaution, justified with an explanation, for the reaction is:

- Use of fume hood
  AND
  SO_2 / sulfur dioxide produced
  AND
  HCl / hydrogen chloride produced; [1 mark]
- Both gases can cause damage to the respiratory tract if inhaled; [1 mark]

[Total: 4 marks]

- You should be able to deduce that the primary alcohol pentan-1-ol reacts with the thionyl chloride
- In this reaction, the -OH group is substituted by a -Cl group
  - Alcohol + thionyl chloride → chloroalkane
  - $CH_3CH_2CH_2CH_2CH_2OH$ (l) + $SOCl_2$ (l) → $CH_3CH_2CH_2CH_2CH_2Cl$ (l)
- You are told that two additional products are produced
  - From what is left, you can piece together that the most likely products are HCl and SO_2
  - $CH_3CH_2CH_2CH_2CH_2OH$ (l) + $SOCl_2$ (l) → $CH_3CH_2CH_2CH_2CH_2Cl$ (l) + HCl (g) + SO_2 (g)
- The HCl and SO_2 products are gases which means the liquid 1-chloropentane can easily be isolated
  - Both HCl and SO_2 are acidic gases which can affect the respiratory system
  - Therefore, the reaction should be completed in a fume cupboard

2a

a) The completed table is:

Organic product	Name	Mechanism involved
Alkene	(2-)methylpropene	Elimination
Alcohol	(2-)methylpropan-2-ol	S_N1

- Correct name and mechanism for the alkene; [1 mark]
- Correct name and mechanism for the alcohol; [1 mark]

[Total: 2 marks]

- When reacted with sodium / potassium hydroxide, halogenoalkanes can form:
  - Alkenes by an elimination reaction
  - Alcohols by a nucleophilic substitution reaction
  - The type and proportion of each product depends on the reaction conditions
    - Generally, alcoholic / ethanolic conditions form the alkene and aqueous conditions for the alcohol
- 2-bromo-2-methylpropane is a tertiary halogenoalkane
  - This means that the nucleophilic substitution mechanism is S_N1

2b

b) The reason for the mixture of products is:

- Conditions / reagents influence the reaction but only to a certain extent
  OR
  Conditions / reagents do not fully influence the reaction; [1 mark]
- The structure of the halogenoalkane will influence the product
  OR
  Tertiary halogenoalkanes tend to favour elimination reactions
  OR
  Primary halogenoalkanes tend to favour substitution reactions; [1 mark]

[Total: 2 marks]

- The structure of a halogenoalkane will influence which product is made
  - Primary halogenoalkanes will tend to favour substitution reactions
  - Tertiary halogenoalkanes will tend to favour elimination reactions
- Refluxing with ethanolic potassium / sodium hydroxide will encourage elimination reactions but the reaction may not be exclusively elimination
- Aqueous potassium / sodium hydroxide will encourage substitution reactions
  - Ethanol can be added to this to increase the solubility of the halogenoalkane in substitution reactions
- You can alter the conditions to influence the ratio of products
  - Though you will still get a mixture of products from elimination and substitution

2c

c) The reaction between butan-2-ol and phosphorus(III) iodide is:

$3CH_3CHOHCH_2CH_3 + PI_3 \rightarrow 3CH_3CHICH_2CH_3 + H_3PO_3$

OR

$3C_4H_{11}O + PI_3 \rightarrow 3C_4H_{10}I + H_3PO_3$

- Correct formulae for reactants and products; [1 mark]
- Correct balancing for reactants and products; [1 mark]

[Total: 2 marks]

- There is enough information here to piece together the equation for this reaction

- Primary halogenoalkanes will tend to favour substitution reactions
- Tertiary halogenoalkanes will tend to favour elimination reactions

- Refluxing with ethanolic potassium / sodium hydroxide will encourage elimination reactions but the reaction may not be exclusively elimination
- Aqueous potassium / sodium hydroxide will encourage substitution reactions
    - Ethanol can be added to this to increase the solubility of the halogenoalkane in substitution reactions
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OR
$3C_4H_{11}O + PI_3 \rightarrow 3C_4H_{10}I + H_3PO_3$

- Correct formulae for reactants and products; [1 mark]
- Correct balancing for reactants and products; [1 mark]

[Total: 2 marks]

- There is enough information here to piece together the equation for this reaction
- Make sure you know the formula for simple molecules such as acids and bases as you may not be given the formula for them in the question
    - If you are not expected to know the formula of an acid or base, such as phosphorous acid, it will be given to you

2d

d) Sulfuric acid can not be used in the preparation of an iodoalkane because:

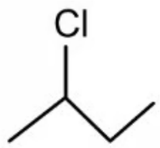
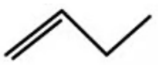
- Sulfuric acid will oxidise I^- ions to I_2 / iodide ions to iodine
  OR
  Iodide ions would be oxidised (as shown in equation) $2I^- \rightarrow I_2 + 2e^-$; [1 mark]

[Total: 1 mark]

- This question links to the reactions of concentrated sulfuric acid with sodium / potassium halides
- These reactions demonstrate the reducing ability of the halide ions
- As you go down the group the reducing ability increases
- In the reaction of concentrated sulfuric acid with sodium iodide, iodine is produced as a redox product
- The concentrated sulfuric acid oxidises the iodide ions to iodine
- The iodide ions reduce the sulfur in sulfuric acid from +6 to -2

3a

a) The skeletal structures of compounds **G** and **H** are:

- Compound **G** =  ; [1 mark]
- Compound **H** =  ; [1 mark]

[Total: 2 marks]

- If the alkene produced does not exhibit geometric isomerism
  - This means that compound **H** cannot be but-2-ene as this forms *E* and *Z* isomers
  - Therefore, compound **H** has to be but-1-ene
  - Cis and trans notation is also accepted

3b

b) The reagents and conditions for reaction **A** are:

- Alcoholic / ethanolic KOH
  OR
  Alcoholic / ethanolic NaOH; [1 mark]
- Heat under reflux; [1 mark]

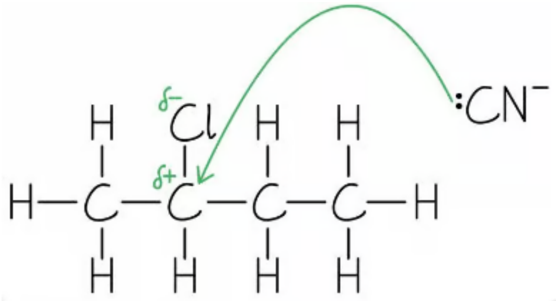
[Total: 2 marks]

- You must know the reagents and conditions for the conversion of a halogenoalkane to an alkene

3c

c) The mechanism for reaction **B** is:

- Nucleophilic substitution / $S_{N;}$ [1 mark]



- Presence of :CN^{-}
  AND
  If the bonding is shown, then the C-N bond must be an unambiguous triple bond; [1 mark]
- Curly arrow from the :CN lone pair to the C-Cl carbon; [1 mark]
- Correct dipole / partial charges on the C-Cl bond
  AND
  Curly arrow from the C-Cl bond to the chlorine atom; [1 mark]

[Total: 4 marks]

- Nucleophilic substitution of halogenoalkanes is an important mechanism to learn
- You are required to know this mechanism where the nucleophile could be
  - The hydroxide ion / :OH^{-}
  - The cyanide ion / :CN^{-}
  - Ammonia / :NH_3
- **Remember:** The curly arrow from the nucleophile must start at the lone pair of electrons
- This is slightly trickier as you do not have a starting point given to you so you must work backwards to the starting compound and then fill in the mechanism

4a

a) The purpose of the condenser is:

Any **one** of the following:

- Returns / condenses volatile reactants / evaporated gases except but-1-ene back to the reaction mixture / so they are not lost; [1 mark]
- Returns 1-bromobutane / water to the reaction mixture / so they are not lost; [1 mark]
- Prevents loss of reactants so they have time to react; [1 mark]
- Allows a higher temperature to be used without loss of reactants; [1 mark]
- Prevents gases other than but-1-ene from entering the gas syringe; [1 mark]

[Total: 1 mark]

- To gain this mark, you need to give more detail than just saying that the condenser cools or condenses the gases
- You would not gain the mark if you said that but-1-ene is condensed as but-1-ene does not condense in the condenser but gaseous but-1-ene will collect in the gas syringe where it will condense when the gas syringe is left to cool

4b

b) The chemical test and expected observation to identify the functional group of the gas in the syringe are:

EITHER

- Bubble the gas through bromine water / aqueous bromine / Br_2 (aq) / bromine in organic solvent; [1 mark]
- Which goes from (red-)brown / orange / yellow to colourless
  OR
  Which decolourises; [1 mark]

OR

- Bubble the gas through acidified potassium manganate(VII); [1 mark]
- Which goes from purple to colourless
  OR
  Which decolourises; [1 mark]

[Total: 2 marks]

- The gas collected in the syringe is but-1-ene
  - Which means that the gas being tested is an alkene with the C=C functional group
- If you describe the test involving bromine water, you would gain a mark to say that bromine water decolourises
  - However, it is best to give the colour change - make sure you state colourless and not clear
- Acidified potassium manganate(VII) is an alternative method which you may not have come across but it is a viable method for distinguishing saturated substances

4c

c) The percentage of 1-bromobutane which was converted to but-1-ene is:

EITHER

- Moles of but-1-ene = $\frac{22}{24\ 000}$ = 9.17 x 10^4 / 9.1667 x 10^4; [1 mark]
- Percentage of 1-bromobutane converted = $\frac{9.1667 \times 10^4}{0.0080}$ x 100 = 11.5 / 11.458%; [1 mark]

OR

- Volume of gas expected = 0.008 x 24 = 0.192 dm^3 **OR** 0.008 x 24 000 = 192 cm^3; [1 mark]
- Percentage of 1-bromobutane converted = $\frac{22}{192}$ x 100 = 11.5 / 11.458%; [1 mark]

[Total: 2 marks]

- A correct answer without working scores 2 marks
- As it is a 1:1 reacting ratio, 0.0080 mol of 1-bromobutane could produce 0.0080 mol but-1-ene
- Use the equation mol = volume of gas (cm^3) ÷ 24 000 to work out the moles of but-1-ene actually produced and compare it as a percentage to the moles of but-1-ene that could have been produced (0.008 mol)
- Or use the equation volume of gas (cm^3) = moles x 24 000 to work out the volume of but-1-ene that could have been produced and then deduce the percentage of the volume actually produced (22.0 cm^3) compared to the theoretical volume
- Whichever method you use, don't forget to convert to a percentage by multiplying by

100

4d

d) The temperature of the gas in the syringe before it cooled is:

- Ratio of volumes before and after cooling = $\frac{24}{22}$ = 1.091 / 1.0909...; [1 mark]
- Temperature of warm syringe = 1.0909 x 298 = 325 K / 325.09090909 K / 52 °C; [1 mark]

[Total: 2 marks]

- The correct answer without working scores 2 marks
- An alternative way to complete this calculation is to use the ideal gas equation:
  - $pV_{before} = nRT_{before}$
    AND
    $pV_{after} = nRT_{after}$
  - Rearranging these gives:
    $\frac{V_{before}}{T_{before}} = \frac{nR}{p}$
    AND
    $\frac{V_{after}}{T_{after}} = \frac{nR}{p}$
- $\frac{nR}{p}$ is a constant value because the pressure and number of moles do not change and R is a constant
  - Therefore:
  - $\frac{V_{before}}{T_{before}} = \frac{V_{after}}{T_{after}}$
  - $\frac{24}{T_{before}} = \frac{22}{298} = 0.07382..$
  - $T_{before} = \frac{24}{0.07382..} = 325.09\text{ K}$

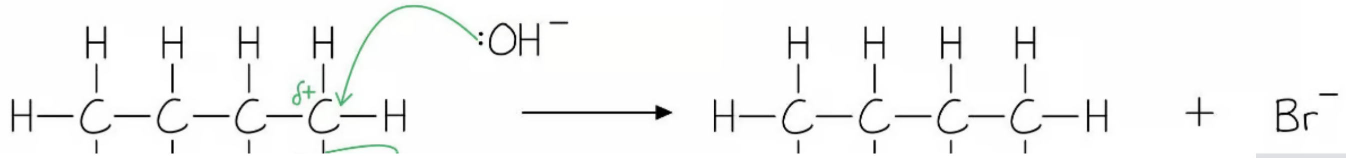
4e

e)

i) The type of reaction taking place to form butan-1-ol is:

- S_N2; [1 mark]

ii) The mechanism for the reaction of 1-bromobutane with hydroxide ions to form butan-1-ol is:



0.07382..

4e

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i) The type of reaction taking place to form butan-1-ol is:

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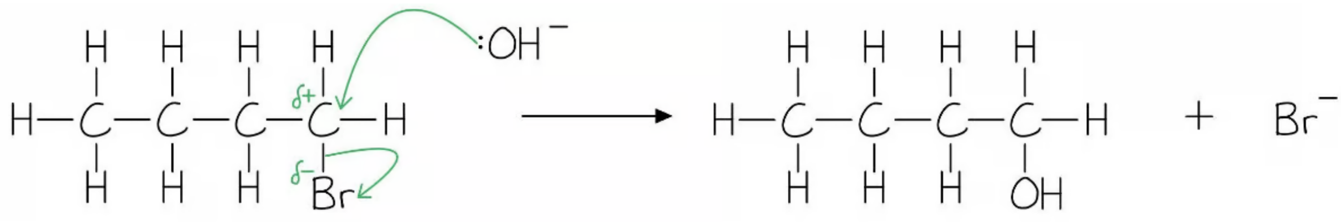


- Presence of $:OH^-$
  AND
  Br^- present as a product; [1 mark]
- Curly arrow from the lone pair of the $:OH^-$ to the carbon of the C-Br bond [1 mark]
- Correct dipole / partial charges on the C-Br bond
  AND
  Curly arrow from the C-Br bond to, or just beyond, the Br atom; [1 mark]

iii) The chemical test and expected observation to confirm the presence of the functional group in butan-1-ol are:

EITHER

- Warm with acidified potassium dichromate(VI) solution; [1 mark]
- Colour / solution changes from orange to green; [1 mark]

OR

- Warm with acidified potassium manganate(VII); [1 mark]
- Which goes from purple to colourless
  OR
  Which decolourises; [1 mark]

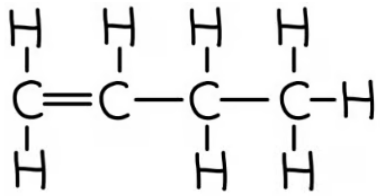
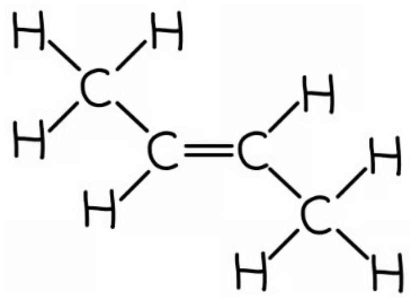
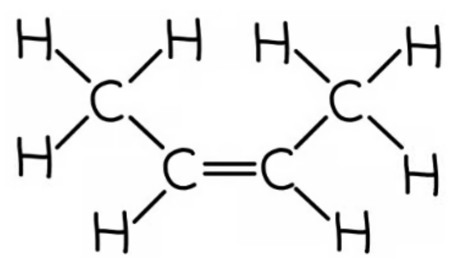
which decolourises; [1 mark]

[Total: 6 marks]

- Alcohols can be formed by reacting a halogenoalkane with an aqueous solution of sodium / potassium hydroxide with ethanol
- The resulting reaction is nucleophilic substitution
  - 1-bromobutane is a primary alcohol which generally undergo S_N2 reactions where the hydroxide ion acts as a nucleophile and substitutes the halogen atom in a one-step reaction
  - You need to state that the reaction is S_N2 to gain the mark
    - Just nucleophilic substitution is not enough for the mark
- You should know the test and observation for a primary alcohol

4f

f) The isomeric alkene products formed in this reaction are:

-  ; [1 mark]
-  ; [1 mark]
-  ; [1 mark]

[Total: 3 marks]

- HBr is eliminated from 2-bromobutane

- The bromine atom is lost from carbon-2
- The hydrogen atom can be lost from carbon-1 or carbon-3
  - The loss of the hydrogen atom from carbon-1 results in but-1-ene
  - The loss of a hydrogen atom from carbon-3 results in the geometrical / cis-trans isomers of but-2-ene